

SOFTWARE REQUIREMENTS SPECIFICATION FOR

TruePixels.rgb An AI-Generated Image Detection System

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1. Introduction

1.1 Purpose

The purpose of this document is to specify, completely and unambiguously, the functional and non-functional requirements of the TruePixels.rgb system. It describes the objectives, features, external interfaces, data requirements, design constraints and quality attributes of the software so that the product can be developed and evaluated against a single, consistent reference. The intended audience comprises the developers who design and implement the system, the project guides and evaluators reviewing it, the testers responsible for verification and validation, future maintainers and end users interested in the system's capabilities.

1.2 Scope

TruePixels.rgb is a web-based application whose primary objective is to determine automatically whether a given still image is Real or AI Generated, using a hybrid framework that combines a CLIP-based semantic classifier with a frequency-artifact classifier. The outputs of both classifiers are fused to produce the final prediction and confidence score. The system shall provide the following capabilities:

- Accept image uploads through a web-based user interface and validate and preprocess them prior to inference;
- Perform semantic feature extraction and frequency-domain analysis on uploaded images;
- Classify each image as Real or AI Generated through decision fusion and report a confidence score;
- Generate an explainability visualization highlighting the regions influencing the prediction;
- Authenticate users and administrators and enforce role-based access control;
- Maintain a per-user prediction history and produce downloadable PDF reports;
- Record system and prediction logs and expose administrative monitoring, analytics, user and model management.

Identification of the specific generative model, detection of localized image editing and deepfake video or other multimedia analysis are explicitly outside the scope of the current release and are identified as candidate enhancements. By providing an accurate and explainable authenticity assessment, the system supports applications in digital forensics, journalism, content moderation and education.

1.3 Definitions, Acronyms and Abbreviations

Term / Acronym	Definition
AI	Artificial Intelligence.
API	Application Programming Interface.
CLIP	Contrastive Language-Image Pre-training; a vision-language model used for semantic image understanding and feature extraction.
Confidence Score	The probability assigned by the hybrid detection framework to its predicted class.

Term / Acronym	Definition
Decision Fusion	Combining the outputs of the semantic and frequency-artifact classifiers to produce the final prediction.
Diffusion Model	A generative model that synthesizes images by progressively removing noise.
Explainability Visualization	A visual representation highlighting the image regions or features that most influenced the model's prediction.
Frequency Artifact Analysis	Detecting synthetic image artifacts in the frequency domain using transforms such as the Fourier Transform.
GAN	Generative Adversarial Network; a class of generative models capable of producing synthetic images.
GUI	Graphical User Interface.
HTTP / HTTPS	Hypertext Transfer Protocol / HTTP Secure.
REST	Representational State Transfer; the architectural style used for the system's web API.
SRS	Software Requirements Specification.
XAI	Explainable Artificial Intelligence.

1.4 References

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- [7] PostgreSQL Documentation, PostgreSQL Global Development Group. <https://www.postgresql.org/docs>

1.5 Overview

The following sections of this document present a detailed description of the system requirements, including both functional and non-functional aspects. It also outlines the interfaces, performance needs, design constraints and safety considerations that guide the development process. Additional sections provide supporting information such as definitions and models that ensure clarity and completeness of the specification.

2. Overall Description

2.1 Product Perspective

TruePixels.rgb is a self-contained, web-based, client-server application for image authenticity verification. A user interacts with a React-based front end, which communicates with a FastAPI back end over HTTPS using a RESTful API.

The back end validates and preprocesses the uploaded image, performs semantic feature extraction using a CLIP-based classifier and frequency-domain analysis using a frequency-artifact classifier, combines the outputs through a decision fusion module to determine the final prediction and confidence score, generates an explainability visualization and persists the relevant records to a PostgreSQL database before returning the results to the client. The overall system architecture is shown in Appendix A (Figure A.1).

2.2 Product Functions

At a high level the system provides the following groups of functions:

- **Account and Access:** Registration, authentication and session management with role-based access control.
- **Image Submission and Analysis:** Upload, validation and preprocessing of still images; semantic feature extraction (CLIP) and frequency-domain artifact analysis; decision fusion producing a Real / AI-Generated label with a confidence score; and explainability visualization.
- **Results and History:** Display of results, per-user prediction history and downloadable PDF reports.
- **Administration:** Monitoring, log review, analytics, user account management and AI model management.

2.3 User Classes

- **General User:** An individual who verifies whether an image is Real or AI Generated. Basic computer and web-browsing skills are assumed but no prior knowledge of machine learning. A User may register, log in, upload images and view prediction results, confidence scores, explainability visualizations and personal prediction history, but has no access to administrative functions, system logs or other users' data.
- **Administrator:** A privileged user responsible for managing and monitoring the system. An Administrator may review prediction and system logs, monitor activity and analytics, manage user accounts and manage the AI model and is responsible for the continued correct operation of the system.

2.4 Constraints

- **C.1 Supported Inputs** — Only still images in JPG, JPEG and PNG formats within the configured maximum file size are accepted.
- **C.2 Binary Classification** — The system labels an image as Real or AI Generated only.

- **C.3 Explainability Availability** — Explainability visualizations are generated only when supported by the deployed inference framework.
- **C.4 Data Dependence** — Detection accuracy depends on the quality, diversity and size of the training data for both classifiers.
- **C.5 Latency Dependence** — Prediction latency depends on the compute available on the deployment server, since both classifiers execute during inference.
- **C.6 Access Control** — Prediction services require authentication and administrative functions require administrator privileges.
- **C.7 Scope Exclusions** — Deepfake video detection, localized editing detection and other multimedia analysis are not supported.
- **C.8 Connectivity** — A stable network connection is required when deployed as a web application and the back end, inference framework and required libraries must be available.

2.5 Assumptions and Dependencies

The requirements rest on the following assumptions:

- Users have access to a modern web browser and a stable network connection;
- Uploaded images are legible and belong to a supported format and users supply valid inputs;
- Administrators hold valid credentials for privileged features;
- Suitable trained semantic and frequency-artifact classifiers have been developed and deployed prior to release.

The system further depends on the following external factors:

- Availability of the FastAPI back end and the hybrid inference framework;
- Availability and integrity of the PostgreSQL database;
- Availability of the supporting libraries PyTorch, Hugging Face Transformers, OpenCV and NumPy;
- Successful execution of the semantic classifier, frequency-artifact classifier and decision fusion module.

3. Specific Requirements

3.1 Functional Requirements

F.1 User Registration

Description: The system shall allow an unregistered visitor to create a user account by submitting the required details, which are validated and persisted. The email must be unique and not already registered; on success a User record with role “User” is created.

Input: Full name; email address (unique); password.

Output: Confirmation of successful account creation, or a descriptive validation or error message.

F.2 User Authentication

Description: The system shall authenticate a registered user by verifying the submitted credentials against stored records and establish a session on success before granting access to protected features.

Input: Email; password.

Output: An authenticated session on success, or an authentication-failure message.

F.3 Administrator Authentication

Description: The system shall authenticate administrators through the administrative login and grant access only to accounts holding the “Admin” role, establishing an elevated-privilege session.

Input: Administrator email; password.

Output: An authenticated administrative session, or an access-denied message.

F.4 Role-Based Access Control

Description: The system shall enforce authorization such that each authenticated principal may access only the functions permitted for its role; unauthorized requests are rejected and, where relevant, logged.

Input: Authenticated session and associated role; requested resource.

Output: Granted or denied access to the requested resource.

F.5 Image Upload

Description: The system shall allow an authenticated user to upload a still image (JPG, JPEG or PNG) for authenticity analysis; the image is recorded in the Image entity and queued for validation.

Input: Image file in JPG, JPEG or PNG format.

Output: Acknowledgement that the image was received for processing, or a rejection message.

F.6 Input Validation

Description: The system shall validate every uploaded image before further processing, verifying file format, file size and file integrity and shall reject corrupted or unsupported inputs so that only valid images proceed to preprocessing.

Input: Uploaded image file.

Output: Validation success, or a descriptive error message identifying the failure.

F.7 Image Preprocessing

Description: The system shall preprocess each validated image by resizing, normalizing pixel values and preparing it for semantic and frequency-domain analysis in accordance with the input specifications of the hybrid inference framework.

Input: Validated image.

Output: A preprocessed image suitable for semantic feature extraction and frequency-domain analysis.

F.8 Image Classification

Description: The system shall classify the preprocessed image as Real or AI Generated by combining the outputs of a CLIP-based semantic classifier and a frequency-artifact classifier through a decision fusion module, which must be deployed and available.

Input: Preprocessed image.

Output: A predicted class label (Real or AI Generated).

F.9 Confidence Score Generation

Description: The system shall compute and report a confidence score for each prediction from the fused outputs of the two classifiers and associate the score with the prediction record.

Input: Outputs of the semantic classifier, frequency-artifact classifier and decision fusion module.

Output: A numeric confidence score for the predicted class.

F.10 Explainability Visualization

Description: The system shall generate an explainability visualization that highlights the image regions or features most influential to the prediction, present it alongside the original image and store it for later retrieval.

Input: Preprocessed image; inference outputs and intermediate model information.

Output: An explainability visualization.

F.11 Result Display

Description: The system shall present the predicted class, confidence score, explainability visualization and appropriate status messages to the user in a single results view.

Input: Prediction result; confidence score; explainability visualization.

Output: A rendered results view.

F.12 Prediction Storage

Description: The system shall persist each prediction together with its associated image reference, model reference, confidence score and timestamp, linked to the corresponding Image, Model and Explainability records.

Input: Prediction result and associated metadata.

Output: A stored prediction record.

F.13 Prediction History

Description: The system shall allow an authenticated user to view the chronological history of predictions associated with their own account only; no other user's records are exposed.

Input: Authenticated user session.

Output: A chronological list of the user's past predictions.

F.14 Report Generation

Description: The system shall, on request, generate a downloadable PDF report of a prediction (outcome, confidence score and explainability visualization) for a prediction owned by the requesting user.

Input: A selected prediction record.

Output: A PDF report file.

F.15 System Logging

Description: The system shall record significant events authentication attempts, prediction requests, administrative actions and errors in the Log entity and make them available to administrators.

Input: System and user events.

Output: Persisted log entries.

F.16 Administrator Dashboard and Monitoring

Description: The system shall provide an administrative dashboard through which authorized administrators may review prediction records, system logs and application activity.

Input: Authenticated administrator session.

Output: Aggregated monitoring views of predictions, logs and activity.

F.17 User Account Management

Description: The system shall allow an administrator to view and to enable, disable or remove user accounts in accordance with policy.

Input: Administrator session; target user account; management action.

Output: Updated account state, or an error message.

F.18 System Analytics

Description: The system shall present administrators with aggregate analytics derived from stored predictions and logs such as prediction counts, class distribution and usage over time without exposing individual credentials.

Input: Stored prediction and log data.

Output: Aggregated analytics views.

F.19 AI Model Management

Description: The system shall allow an authorized administrator to upload or update the semantic classifier, frequency-artifact classifier or decision fusion configuration and activate a selected model version for subsequent predictions.

Input: Model artifact; model metadata (name, version, training reference); activation action.

Output: A registered or updated model record and confirmation of the active model configuration.

F.20 Error Handling

Description: The system shall detect and handle error conditions gracefully invalid credentials, unsupported or corrupted files and inference or server failures returning meaningful messages without disclosing sensitive internal details, remaining stable and logging the event.

Input: An error-producing condition.

Output: A user-appropriate error message and a corresponding log entry.

3.2 External Interface Requirements

3.2.1 User Interfaces

Description:

The system shall provide a responsive, web-based graphical user interface with role-based access. The User interface shall support account registration and login, image upload, input-validation feedback, display of the predicted class and confidence score, display of the explainability visualization alongside the original image, access to personal prediction history and clear success and error messages. The Administrator interface shall provide a secure login and a dashboard for monitoring activity, reviewing prediction and system logs, viewing analytics, managing user accounts and managing the AI model.

3.2.2 Hardware Interfaces

Description:

The client requires standard hardware capable of running a modern web browser. The server shall meet at least a minimum configuration of an Intel Core i3 (or equivalent) processor, 4 GB RAM and 2 GB free storage; an Intel Core i5 / AMD Ryzen 5 or higher with 8 GB RAM and 5 GB free storage is recommended. A CUDA-enabled NVIDIA GPU is recommended for model training and for accelerated inference where available.

3.2.3 Software Interfaces

Description:

The front end is implemented in React and the back end in FastAPI (Python 3.x). The hybrid inference framework and explainability generation are implemented using PyTorch and Hugging Face Transformers; the CLIP-based semantic classifier is loaded through Hugging Face Transformers. Image preprocessing and frequency-domain operations use OpenCV and NumPy and persistence uses PostgreSQL. The client runtime is a modern web browser and the server components run on a Windows, Linux or macOS host.

3.2.4 Communications Interfaces

Description:

The front end and back end shall communicate through RESTful API endpoints over HTTP and over HTTPS when deployed. Each request shall be authenticated where required; the back end shall validate file type, size and integrity before processing; and the API response shall convey the prediction, confidence score and explainability visualization. Passwords shall be stored using secure hashing and access to prediction logs, user information and administrative data shall be restricted to authorized principals.

3.3 Other Non-Functional Requirements

NF.1 Response Time

Description:

Under normal operating conditions the system shall validate, preprocess, execute both classifiers, fuse the outputs, generate the explainability visualization and present results within a response time suitable for interactive use, verified during system testing.

NF.2 Capacity

Description:

The system shall support multiple concurrent analysis requests under normal load without significant degradation and the architecture shall accommodate future scaling.

NF.3 Accuracy

Description:

The hybrid framework shall be evaluated using Accuracy, Precision, Recall, F1-Score and ROC-AUC on an independent test dataset; final figures are established during system validation.

NF.4 Resource Utilization

Description:

The system shall use CPU and memory efficiently during inference, employ GPU acceleration where available and perform database operations without perceptible delay.

NF.5 Reliability

Description:

The system shall produce consistent predictions for identical inputs, handle invalid inputs and runtime exceptions without unexpected failure and persist records reliably.

NF.6 Availability**Description:**

The system shall be available whenever the back-end server is operational, with scheduled maintenance downtime minimized.

NF.7 Security**Description:**

The system shall require authentication before protected features, enforce role-based authorization, store passwords as secure hashes, validate uploaded files, restrict sensitive data to authorized administrators and use HTTPS when deployed.

NF.8 Maintainability**Description:**

The system shall follow a modular architecture in which the authentication, inference, explainability, logging and persistence modules are independently maintainable, with documented source code.

NF.9 Portability**Description:**

The system shall be deployable on Windows, Linux and macOS and on both local and cloud servers with minimal configuration change.

NF.10 Scalability**Description:**

The architecture shall support multiple concurrent users and permit future expansion, including additional classifiers, fusion strategies and higher user loads.

NF.11 Usability**Description:**

The interface shall be intuitive, error messages shall be clear and informative and explainability visualizations shall aid interpretation of predictions.

NF.12 Extensibility**Description:**

The system shall allow future integration of additional classifiers, fusion strategies and explainability techniques and enhancements such as video detection, batch processing and multi-class identification, with minimal architectural change.

NF.13 Interpretability**Description:**

The system shall provide an explainability visualization for each prediction to improve transparency and user confidence.

3.4 Constraints

Technology: The front end shall use React; the back end FastAPI; the inference framework PyTorch with Hugging Face Transformers; preprocessing OpenCV and NumPy; and persistence PostgreSQL.

Model: Binary classification only (Real or AI Generated) using a hybrid CLIP semantic classifier and frequency-artifact classifier combined through decision fusion; explainability through framework-supported visual techniques; still images only.

Security: Authentication is required before prediction services; administrative features are restricted to administrators; passwords are stored as secure hashes; and uploaded files are validated before processing.

Data: Only JPG, JPEG and PNG images within the configured maximum file size are accepted; records and logs are stored securely; and images may be stored as file references rather than binary content in the database.

Deployment: The server shall have Python and the required machine-learning libraries installed; the back end and inference framework must be operational for prediction requests; and a stable network connection is required when deployed online.

Appendix A

A.1 System Architecture

The system architecture in Figure A.1 shows the React front end communicating with the FastAPI back end over HTTPS. The back end invokes the hybrid inference engine, which comprises the CLIP-based semantic classifier and the frequency-artifact classifier feeding a decision fusion module; the fused result yields the prediction with a confidence score and drives the explainability module. Prediction records and logs are persisted to the PostgreSQL database.

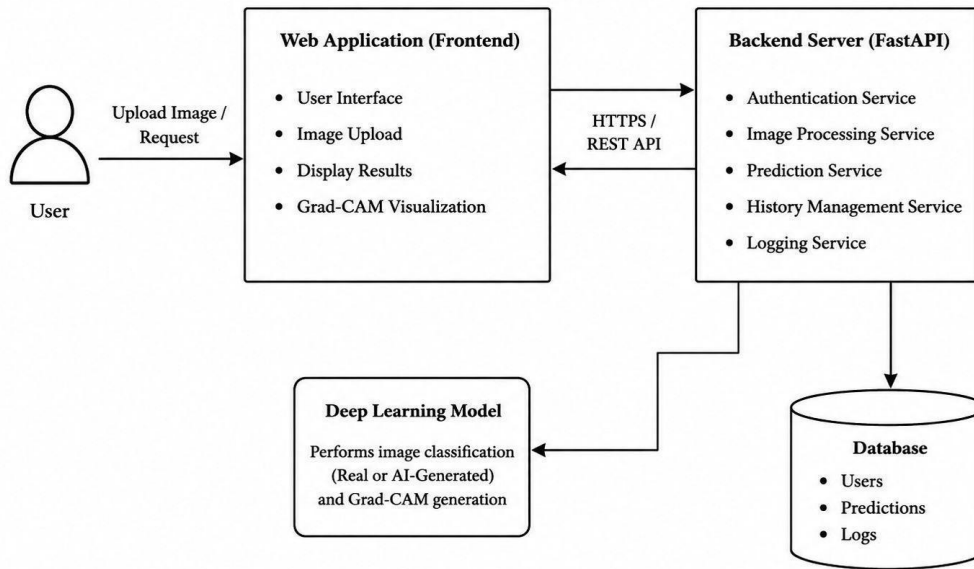


Figure A.1 System architecture of TruePixels.rgb

A.2 Operational Workflow

The operational workflow in Figure A.2 traces a request end to end: a principal accesses the application and registers or logs in, an authenticated user uploads an image and the system validates it, rejecting invalid images with an error message. Valid images are preprocessed, then passed through semantic feature extraction (CLIP) and frequency artifact analysis; the outputs are combined by decision fusion, an explainability visualization is generated and the result is stored and displayed. The user may optionally download a PDF report.

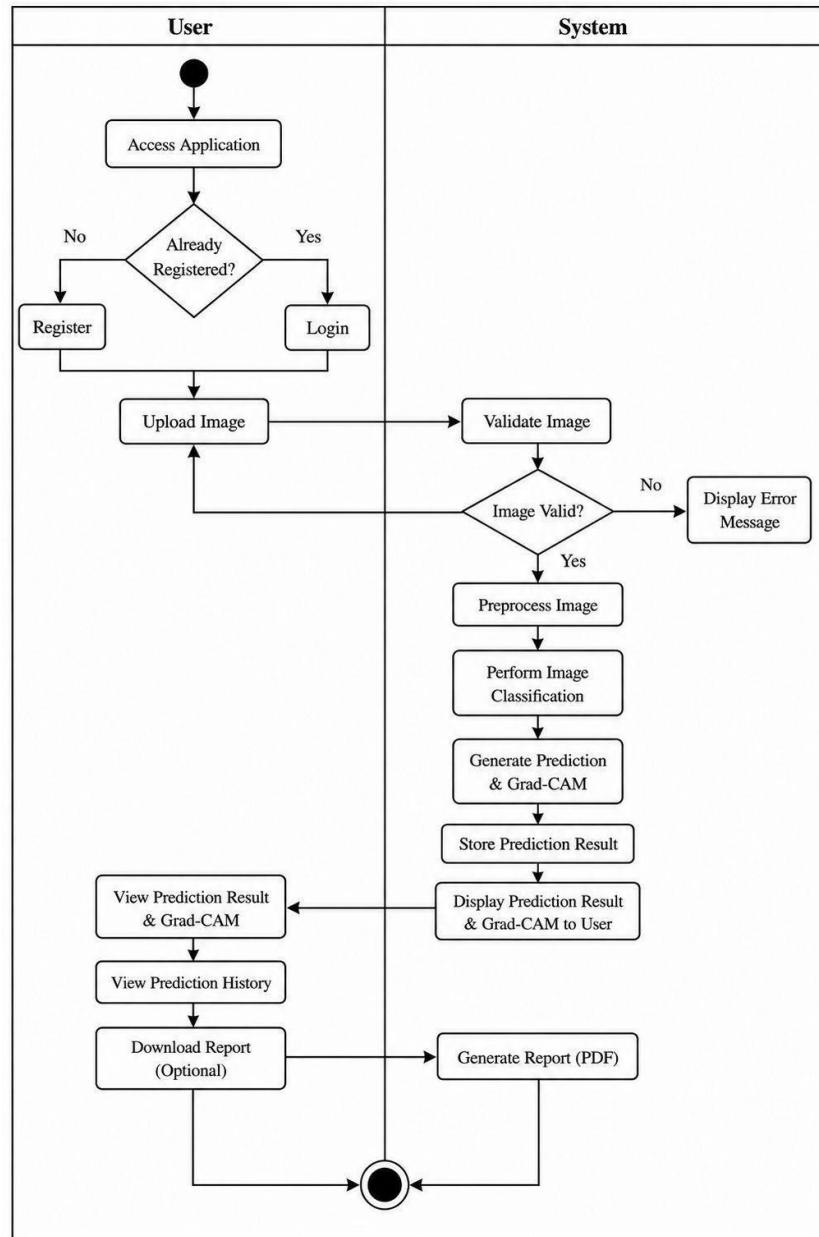


Figure A.2 — Operational workflow of TruePixels.rgb

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A

AI — Artificial Intelligence.

API — Application Programming Interface.

C

CLIP — Contrastive Language-Image Pre-training; semantic feature extractor.

Confidence Score — Probability assigned to the predicted class.

CUDA — Compute Unified Device Architecture; NVIDIA GPU acceleration.

D

Decision Fusion — Combining classifier outputs into a single prediction.

Diffusion Model — Generative model that synthesizes images by denoising.

E

Explainability Visualization — Visual map of regions influencing a prediction.

F

Frequency Artifact Analysis — Spectral detection of synthetic image artifacts.

G

GAN — Generative Adversarial Network.

GUI — Graphical User Interface.

H

HTTP / HTTPS — Hypertext Transfer Protocol / HTTP Secure.

R

Real / AI Generated — The two output classes of the detector.

REST — Representational State Transfer web-API style.

S

SRS — Software Requirements Specification.

X

XAI — Explainable Artificial Intelligence.